

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SP9 - Enterprise Risk Management

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** A financial services company was founded by a group of consumer activists who envisaged a community-driven bank in a particular country. The objective was to support the community's capital requirement for small and medium enterprises. The bank has expanded its operations to 12 branches in the region.

The company has been financially strong, but its customer base is predominantly high-age. The company is concerned about fraud and cyber incidents in the region, which have also affected it lately. Additionally, the country's regulator has amended the relevant regulatory provisions pertaining to anti-money laundering. The company is starting to review its business in order to comply with the new regulatory norms.

The company's board has agreed to incorporate an Enterprise Risk Management (ERM) program, which it hopes will help address a wide range of risks, including financial, societal, and technological risks.

You are a senior-level executive in the company from Finance with rich experience in risk management and leadership qualities. The Board has selected you to lead the company's ERM initiative.

Your first meeting with the Board was held, in which the fundamentals regarding the ERM initiative were discussed and principally agreed upon.

- i)** Discuss some of the guiding principles and their advantage for implementing the ERM at the company in the given context. (15)

The Board has asked for a knowledge-sharing (training) session for the board members which may lead to approval for implementation of ERM in the company. The session will cover ERM and risk assessment areas.

- ii)** Outline the topics which can be covered in the training session. (3)

- iii)** What are the key financial, societal, technological, and regulatory risks that a community-driven financial services company may face? (7)

The training session was held, and it helped bring the Board members to the same level of understanding, expectations, and outcomes of the ERM process. The Board also subsequently approved the implementation of ERM. As a next step, you have decided to draft a risk register for the company.

- iv)** Design a risk-register, in the given context and risk-profile of the company. (10)

[35 Marks]

- Q. 2)** You are the Head of the catastrophe risk modelling team in a large reinsurance company. You have led a team to develop a robust model for cyclones and are going to demonstrate the use of the model for estimation of modelled losses to a medium-scale manufacturing unit, from cyclone in a particular location.

- i)** Outline the illustrative steps a user will typically perform when using the Catastrophe model. (6)

In the demonstration session of cyclone catastrophe model to the users and management, a participant raised a discussion point on the data used to build the model and quality-related challenges therein.

- ii) Discuss the role of data quality in the development and use of catastrophe models, particularly for cyclone-related risk estimation in industrial settings. (8)

The company is working on capital allocation based on economic capital (EC) approach. You are considering two additional risk-adjusted performance measurement indicators other than Return on Equity (ROE) and Return on Assets (ROA) – Risk-adjusted return on capital (RAROC) and Risk-adjusted return on risk-adjusted capital (RAROROC). The RARORAC is defined as Risk-adjusted return / required-capital i.e. (revenue – costs – expected losses) / required capital.

The EC approach is used as following:

L: largest liability portfolio that portfolio of assets can support over a 1-year horizon (while in a going-concern basis)

EL = extreme loss corresponding to a 1-in 300 year risk event

r_a = return on the assets' portfolio in the year

r_l = increase in liabilities over the year

Liability is given by Asset times $\{(1 + r_a) * (1 - EL) / (1 + r_l)\}$

Consider the above parameters and EC as the difference between assets and liabilities.

Total Assets of 150, comprise of only Assets 'A' and 'B', with the percentage of A in the portfolio = 65%.

Return of A = 7.29%

Return of B = 7.11%

Fall in A = 21%

Fall in B = 12%

Correlation (r) between A and B = 0.5

Increase in liabilities over the year = 5.32%

Combined fall in the portfolio (with the diversification effect) = 14.40%

You are working out the outcomes of a pro-rata capital allocation approach and want to analyse the RARORAC for A and B.

- iii) Discuss the utility of ROE, ROA, RAROC and RARORAC for the purpose as risk performance measures. (7)
- iv) Calculate the impact of diversification effect of the assets on the economic capital as a percentage changes in the EC with and without diversification. (3)
- v) Prepare a summary of the balance sheet at the start of the year for each asset and total with asset value, liabilities and EC. (8)
- vi) Prepare a summary of the balance at the end of the year. Compare the RARORAC of each of A and B and comment. Assume that the impact on asset value is multiplicative w.r.t. growth (at the given rates) and expected losses (for A and B at 1.45% and 1.26% respectively, taking into account diversification effect and over 1-year time horizon). (8)

[40 Marks]

- Q. 3) You are an Actuary and the Chief Risk Officer (CRO) of a large general insurance company.

The company is exploring opportunities in the emerging cyber insurance market for smart homes. With the increasing adoption of internet of things (IoT) devices, homeowners face rising cyber threats, including unauthorized access, data breaches, ransomware attacks, etc.

Incorporating the systemic and complex dependencies with such risks would require a robust actuarial framework to quantify these risks, price policies appropriately, and ensure long-term portfolio sustainability.

The company already has an ERM in place. The CEO has asked you to provide input to the Chief Actuary and Pricing Actuary in consideration of the above opportunity of cyber insurance for smart homes.

- i) Discuss a robust and comprehensive framework for risk measurement and assessment for cyber insurance for smart homes. (7)

The company has a robust mechanism for model review processes with multiple layers of testing and lines of defence, before its outputs are published for various purposes such as for shareholder' information, internal management, regulatory reporting etc.

A Head of Actuarial Modelling has joined the Actuarial team in your company, whose prior experience is in a team where the models are tested comprehensively at one line of defence only. The Head has communicated and suggested to you and senior management that multiple layers of defence of model review be removed as they add unnecessary overhead and make the process inefficient, and instead has proposed more resources and efforts on the 1st and only line of defence for such model reviews.

- ii) Draft the content of the response to the stakeholders (the Head of Actuarial modelling, senior management, etc.) to reinforce your support in continuing such existing model review processes. (7)

The team leader of the model change management was leading task of deploying a set of 15 model changes which included fixes to some errors and updates on some work flow. The estimated impact of such changes on the reserves was a decrease of Rs 89 Cr overall. As per company's policy, the team leader has the autonomy to decide whether the set of changes requires an independent review or not, if the impact of changes is below Rs 100 Cr. . The deployment has had timeline of completion by end of this quarter for which only 2 weeks are left. The team leader takes judgement call to not go ahead with the independent review and goes ahead with the deploying the changes into production. The resultant reserves are to be utilised for year-end reporting.

Auditors found that 4 out of those 15 model changes were not implemented suitably and had offsetting impact in the estimated impact.

- iii) How would you mitigate the likelihood of such a risk and how would you justify the importance of robust model change management in spite of continuous pressure to improve models within tight deadlines? (6)

The team leader of a model production team led a project on deploying a new model for valuation of a certain line of business and completed the task well within the timeline of the year-end regulatory period. Just before 20 days before the end of financial year, the model development team discovered that assumptions for two products under that line of business have been interchanged and this results in incorrect reserves. As the two products are low on materiality in terms of premiums and reserves, they were not included in the model validation review.

- iv) How would you (re) design your model validation framework to effectively mitigate such a risk. (5)

[25 Marks]
